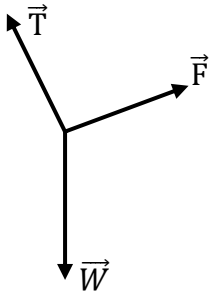
Physics 1 Problem Solving Video - Vectors

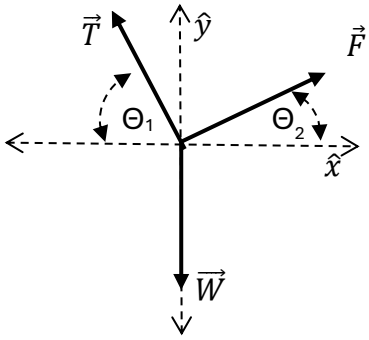
Three force vectors are acting on an object and are shown below, the sum of the vectors acting on the object is given by the sum of the vectors and is the net force $\Sigma\vec{F}$.

$$\Sigma\vec{F} = \vec{T} + \vec{F} + \vec{W}$$

Vectors acting on an object



Vectors shown in coordinate system



A) Determine the magnitude and direction of the net force acting on the object given the magnitude and directions of the vectors below.

$|\vec{T}| = 200\ N$

$|\vec{F}| = 50\ N$

$|\vec{w}| = 300\ N$

$\Theta_1 = 60°$

$\Theta_2 = 20°$

Answer: $|\Sigma\vec{F}|$ = 121.8 N α = 116° and β = 153°

B) If only the magnitude of the forces change on the object above so that the net force acting on the object is equal to zero, determine the magnitude of the $|\vec{W}|$ vector in terms of the magnitude of the $|\vec{F}|$ and $|\vec{T}|$vectors and the angles Θ_1 and Θ_2 only.

$$\Sigma\vec{F} = \vec{T} + \vec{F} + \vec{W} = 0N$$

Answer: $|\vec{w}| = |\vec{T}|\cos(90° - \Theta_1) + |\vec{F}|\cos(90° - \Theta_2)$

C) From the equation derived in Part B solve for the magnitude of the $|\vec{F}|$ vector.

Answer: $|\vec{F}| = \frac{|\vec{w}| - |\vec{T}|\cos(90° - \Theta_1)}{\cos(90° - \Theta_2)} = 1248\ N$